

RL Policy Gradient Derivation

Let $R(\tau) = \sum_{t=0}^{T-1} \gamma^t r_t$, where $R(\tau)$ is the total reward from taking trajectory τ , where
 $\tau = [s_0, a_0, s_1, a_1, \dots, s_{T-2}, a_{T-2}, s_{T-1}]$ and $a_t = \pi_\theta(a_t | s_t)$,
 and r_t is the reward of taking a_t at s_t .
 (START) probability that our policy chooses action t at state t .

Environment:

- $p(s_0)$: Initial state distribution
- $p(s_{t+1} | s_t, a_t)$: transition probability
state

A good evaluation of how good our policy is

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} [R(\tau)] \approx \sum_{\tau} \underbrace{P(\tau | \pi_\theta)}_{\text{probability of the entire trajectory under policy } \pi_\theta} R(\tau)$$

$$P(\tau | \pi_\theta) = P(s_{T-1}, a_{T-2}, s_{T-2}, \dots, s_1, a_0, s_0 | \pi_\theta)$$

$$P(A \cap B | C) = P(A | B \cap C) P(B | C)$$

$$= P(s_{T-1}, a_{T-2}, s_{T-2}, \dots, s_1, a_0 | s_0, \pi_\theta) P(s_0 | \pi_\theta) \quad \text{independence}$$

$$= P(s_{T-1}, a_{T-2}, s_{T-2}, \dots, s_1 | a_0, s_0, \pi_\theta) P(a_0 | s_0, \pi_\theta) P(s_0)$$

$$= P(s_{T-1}, a_{T-2}, s_{T-2}, \dots | s_1, a_0, s_0, \pi_\theta) P(s_1 | s_0, a_0, \pi_\theta) \underbrace{\pi_\theta(a_0 | s_0)}_{\text{independence}} P(s_0)$$

$$= P(s_{T-1}, a_{T-2} | s_{T-2}, \dots, s_1, a_0, s_0, \pi_\theta) P(s_{T-2} | s_{T-3}, a_{T-3}) \dots P(s_1 | s_0, a_0) \pi_\theta(a_0 | s_0) P(s_0)$$

$$= P(s_{T-1} | a_{T-2}, s_{T-2}, \dots, s_1, a_0, s_0, \pi_\theta) \pi_\theta(a_{T-2} | s_{T-2}) P(s_{T-2} | s_{T-3}, a_{T-3}) \dots P(s_1 | s_0, a_0) \pi_\theta(a_0 | s_0) P(s_0)$$

$$= P(s_{T-1} | s_{T-2}, a_{T-2}) \dots$$

$$= P(s_0) \prod_{t=0}^{T-2} \pi_\theta(a_{t+1} | s_t) P(s_{t+1} | s_t, a_t)$$

policy parameters
(Ex. MLP weights & biases).

we want $\nabla_\theta J(\theta) = \nabla_\theta \sum_{\tau} P(\tau | \pi_\theta) R(\tau)$ where $\nabla_\theta J(\theta)$ denotes direction to change θ to maximize $J(\theta)$

$$= \sum_{\tau} R(\tau) \nabla_\theta P(\tau | \pi_\theta)$$

$$\text{log rule: } \nabla_x \log(f(x)) = \frac{1}{f(x)} \cdot \nabla_x f(x)$$

$$= \sum_{\tau} R(\tau) P(\tau | \pi_\theta) \nabla_\theta (\log(P(\tau | \pi_\theta)))$$

$$\Leftrightarrow \nabla_x f(x) = f(x) \nabla_x \log(f(x))$$

$$\text{so, } \nabla_\theta P(\tau | \pi_\theta) = P(\tau | \pi_\theta) \nabla_\theta \log(P(\tau | \pi_\theta))$$

$$= \sum_{\tau} P(\tau | \pi_\theta) [R(\tau) \nabla_\theta \log(P(\tau | \pi_\theta))]$$

$$= \mathbb{E}_{\tau \sim \pi_\theta} [R(\tau) \nabla_\theta \log(P(\tau | \pi_\theta))]$$

best direction = average best direction to maximize each trajectory weighted by that trajectory's reward.

To improve expected reward, sample trajectories, see how good they were, and adjust params towards making good outcomes more likely.

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$$E_{\tau \sim \pi_\theta} [R(\tau) \nabla_\theta \log(p(\tau|\pi_\theta))]$$

Remember $\log(\prod_i a_i) = \sum_i \log(a_i)$

$$\log(p(\tau|\pi_\theta)) = \log\left(p(s_0) \prod_{t=0}^{T-2} \pi_\theta(a_t|s_t) p(s_{t+1}|s_t, a_t)\right)$$

$$= \log(p(s_0)) + \sum_{t=0}^{T-2} \log(\pi_\theta(a_t|s_t)) + \sum_{t=0}^{T-2} \log(p(s_{t+1}|s_t, a_t))$$

$$\nabla_\theta \log(p(\tau|\pi_\theta)) = \nabla_\theta \log(p(s_0)) + \sum_{t=0}^{T-2} \nabla_\theta \log(\pi_\theta(a_t|s_t)) + \sum_{t=0}^{T-2} \nabla_\theta \log(p(s_{t+1}|s_t, a_t))$$

$$= 0 + \sum_{t=0}^{T-2} \nabla_\theta \log(\pi_\theta(a_t|s_t)) + 0$$

$$\text{so, } \nabla_\theta J(\theta) = E_{\tau \sim \pi_\theta} \left[R(\tau) \sum_{t=0}^{T-2} \nabla_\theta \log(\pi_\theta(a_t|s_t)) \right] \quad \text{where } R(t) = \sum_{k=0}^{T-1} \gamma^k r_k$$

$$= E_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{T-2} (r_0 + \gamma r_1 + \gamma^2 r_2 + \dots) \nabla_\theta \log(\pi_\theta(a_t|s_t)) \right]$$

$$\text{By LOE} = \sum_{t=0}^{T-2} E_{\tau \sim \pi_\theta} \left[(r_0 + \gamma r_1 + \gamma^2 r_2 + \dots) \nabla_\theta \log(\pi_\theta(a_t|s_t)) \right] \quad \begin{array}{l} \text{define } r_0 + \dots + \gamma^{t-1} r_{t-1} = R_{\text{past}} \\ \text{and } G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} \\ \text{so } R(t) = R_{\text{past}} + \gamma^t G_t \end{array}$$

$$\text{By LOE} = \sum_{t=0}^{T-2} E_{\tau \sim \pi_\theta} \left[\underbrace{R_{\text{past}}}_{\text{consists of } r_0, \dots, r_{t-1}} \nabla_\theta \log(\pi_\theta(a_t|s_t)) \right] + E_{\tau \sim \pi_\theta} \left[\gamma^t G_t \nabla_\theta \log(\pi_\theta(a_t|s_t)) \right]$$

and since we are up to time t and know s_t ,
we know $s_0, \dots, s_t, a_0, \dots, a_{t-1}$, so R_{past} is a constant.

$$= \sum_{t=0}^{T-2} R_{\text{past}} E_{a \sim \pi_\theta} \left[\nabla_\theta \log(\pi_\theta(a_t|s_t)) \right] + E_{\tau \sim \pi_\theta} \left[\gamma^t G_t \nabla_\theta \log(\pi_\theta(a_t|s_t)) \right]$$

$$= \sum_{t=0}^{T-2} R_{\text{past}} \sum_a \underbrace{\pi_\theta(a_t|s_t) \nabla_\theta \log(\pi_\theta(a_t|s_t))}_{\substack{\downarrow \text{log rule} \\ \nabla_\theta \pi_\theta(a_t|s_t)}} + E_{\tau \sim \pi_\theta} \left[\gamma^t G_t \nabla_\theta \log(\pi_\theta(a_t|s_t)) \right]$$

$$= \sum_{t=0}^{T-2} R_{\text{past}} \sum_a \nabla_\theta \pi_\theta(a_t|s_t) + E_{\tau \sim \pi_\theta} \left[\gamma^t G_t \nabla_\theta \log(\pi_\theta(a_t|s_t)) \right]$$

$$= \sum_{t=0}^{T-2} R_{\text{past}} \underbrace{\nabla_\theta \sum_a \pi_\theta(a_t|s_t)}_{=1=0} + E_{\tau \sim \pi_\theta} \left[\gamma^t G_t \nabla_\theta \log(\pi_\theta(a_t|s_t)) \right]$$

$$= \sum_{t=0}^{T-2} E_{\tau \sim \pi_\theta} \left[\gamma^t G_t \nabla_\theta \log(\pi_\theta(a_t|s_t)) \right] = E_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{T-2} \gamma^t G_t \nabla_\theta (\log(\pi_\theta(a_t|s_t))) \right]$$

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We just derived that $\nabla_{\theta} J(\theta) = \mathbb{E}_{T \sim \pi_{\theta}} \left[\sum_{t=0}^{T-2} \gamma^t G_t \nabla_{\theta} \log(\pi_{\theta}(a_t | s_t)) \right]$

To improve expected reward, sample trajectories, evaluate how good they were at each timestep, and move in directions that make good outcomes more likely.

How? Sample N trajectories. By LLN, $\mathbb{E}[X] \approx \frac{1}{N} \sum_{i=1}^N X_i$.

$$\text{So, } \nabla_{\theta} J(\theta) \approx \frac{1}{N} \sum_{i=1}^N \sum_{t=0}^{T_i-2} \gamma^t G_t^i \nabla_{\theta} (\log(\pi_{\theta}(a_t^i | s_t^i))) \leftarrow \text{approximation}$$

Implementation

Start with $N=1$. $\nabla_{\theta} J(\theta) \approx \sum_{t=0}^{T-2} \gamma^t G_t \nabla_{\theta} (\log(\pi_{\theta}(a_t | s_t)))$

We want $J(\theta) = J(\theta) + \eta \nabla_{\theta} J(\theta)$ to maximize expected reward.

But pytorch does $J(\theta) = J(\theta) - \eta \nabla_{\theta} L(\theta)$.

We want to find loss function $L(\theta)$ st

$$-\nabla_{\theta} L(\theta) = \nabla_{\theta} J(\theta)$$

$$\nabla_{\theta} L(\theta) = -\nabla_{\theta} J(\theta)$$

$$= -\sum_{t=0}^{T-2} \gamma^t G_t \nabla_{\theta} \log(\pi_{\theta}(a_t | s_t))$$

$$\text{So, } L(\theta) = -\sum_{t=0}^{T-2} \gamma^t G_t \log(\pi_{\theta}(a_t | s_t))$$

compute G_t backward from the sampled rewards and treat G_t as constant during backprop.